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5.5 ROTATIONAL DYNAMICS

The relations among torque, angular acceleration, angular momentum, angular velocity and moment of inertia were seen in Section 5.2. In continuation to that, in this section, we will learn the relations among the other dynamical quantities like work, kinetic energy in rotational motion of rigid bodies. Finally a comparison between the translational and rotational quantities is made with a tabulation.

5.5.1 Effect of Torque on Rigid Bodies

A rigid body which has non zero external torque (τ) about the axis of rotation would have an angular acceleration (α) about that axis. The scalar relation between the torque and angular acceleration is,

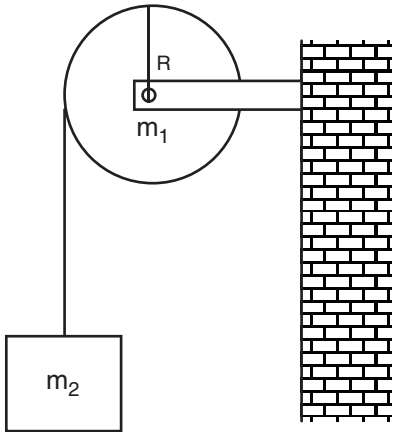
$$\tau = I\alpha \qquad (5.48)$$

where, I is the moment of inertia of the rigid body. The torque in rotational motion is equivalent to the force in linear motion.

EXAMPLE 5.18

A disc of mass 500 g and radius 10 cm can freely rotate about a fixed axis as shown in figure. light and inextensible string is wound several turns around it and 100 g body is suspended at its free end. Find the acceleration of this mass. [Given: The string makes the disc to rotate and does not slip over it. $g = 10\ m\,s^{-2}$.]

Solution



Let the mass of the disc be m_1 and its radius R. The mass of the suspended body is m_2.

$m_1 = 500\ g = 500\times10^{-3}\ kg = 0.5\ kg$
$m_2 = 100\ g = 100\times10^{-3}\ kg = 0.1\ kg$
$R = 10\ cm = 10\times10^{-2}\ m = 0.1\ m$

As the light inextensible string is wound around the disc several times it makes the disc rotate without slipping over it. The translational acceleration of m_2 and tangential acceleration of m_1 will be the same. Let us draw the free body diagram (*FBD*) of m_1 and m_2 separately.

FBD of the disc:

